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Objective: Set up the phase 4 repo and update graph.h with versioned edges and shared_mutex.

Attempts made:

Started by copying phase 3 as the base since ConcurrentQueue carries over unchanged. The first thing I needed to figure out was how to make queries deterministic when updaters are running at the same time. The answer is versioning — every edge gets tagged with the version of the ADD that created it, and a query at version q just skips any edge with version $\geq q$. That way it doesn't matter if an updater has already appended newer edges by the time the SSSP scan runs.

The other change was switching Graph's lock vector from mutex to shared_mutex. My first instinct was to keep regular mutexes but that would serialize the SSSP scan — if TQ threads all need to read the same vertex's adjacency list, only one could go at a time. shared_mutex lets all readers in simultaneously and only blocks for an exclusive writer.

Solutions/Workarounds:

No issues this session. Confirmed Makefile already had -std=c++20 which shared_mutex needs.

Commits: add versioned edges and query request types to graph.h

Experiments/Analysis:

No experiments this session.

Learning outcome:

shared_mutex is the right choice when you have many concurrent readers and infrequent writers. Per-vertex locks mean updaters and SSSP workers on different vertices never interfere with each other at all.